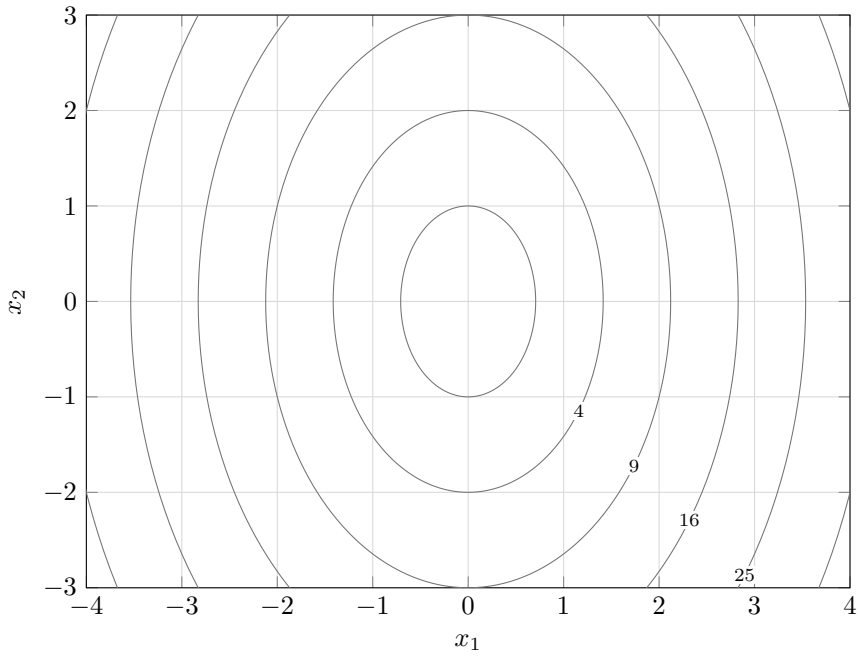
ASEN 3502 Computational Methods
Exam 3 Practice Problems (2026)

Instructions: These problems are meant to show the style of questions on Exam 3; they are not an exhaustive list of what may be asked. These problems were not timed, so they may not be indicative of the real exam length.

Question 1. (30 points) Consider the unconstrained optimization problem of minimizing $f(\vec{x}) = 2x_1^2 + x_2^2$, starting from $\vec{x}_0 = [1,\ 2]^T$. Contours of f are plotted below.



(a) Compute the gradient descent search direction $\vec{p}_0$ at $\vec{x}_0$.

(b) Mark $\vec{x}_0$ on the contour plot and draw a line from it along the search direction.

(c) Mark point A, reached with step size $\alpha = 1$, and point B, reached with $\alpha = 0.5$.

(d) Which of the points A and B satisfy the Armijo condition with sufficient decrease factor $\mu_1 = 10^{-4}$? Show your work.

Question 2. (20 points) The Matlab function below implements Adam, but the line that updates `x` is incomplete.

```
function x = adam(grad_f, x0, alpha, gamma_v, gamma_s, epsilon, tol)
    x = x0;
    v = zeros(size(x0));
    s = zeros(size(x0));
    k = 0;
    while norm(grad_f(x)) > tol
        g = grad_f(x);
        v = gamma_v*v + (1 - gamma_v)*g;
        s = gamma_s*s + (1 - gamma_s)*g.^2;
        v_hat = v/(1 - gamma_v^(k+1));
        s_hat = s/(1 - gamma_s^(k+1));

        % TODO: implement x update below

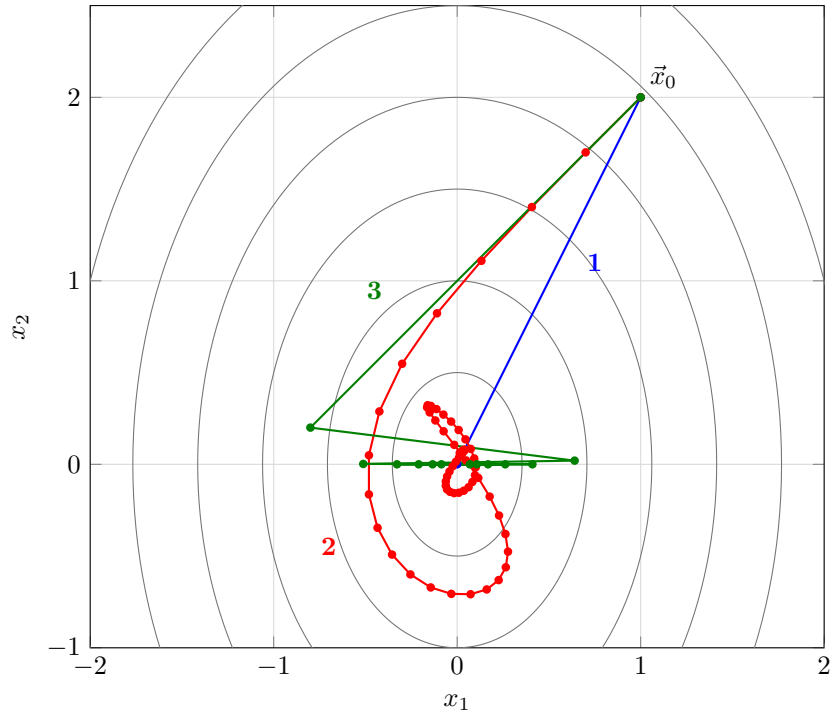
        x =

        k = k + 1;
    end
end
```

(a) Complete the line that updates `x`.

(b) Write a Matlab script that uses this function to minimize $f(\vec{x}) = 2x_1^2 + x_2^2$ starting from $\vec{x}_0 = [1,\ 2]^T$ with $\alpha = 0.3$, $\gamma_v = 0.9$, $\gamma_s = 0.999$, $\epsilon = 10^{-8}$ and a tolerance of 10^{-6}.

(c) The plot below shows the paths taken by Adam, Newton's method, and gradient descent with a fixed step size, all starting from $\vec{x}_0$. Which path (1, 2, or 3) will your script produce? Justify your answer.

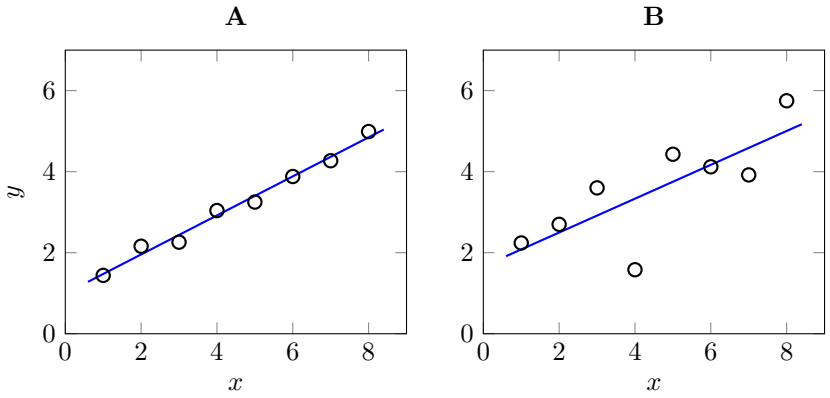


Question 3. (20 points) We want to find the $\vec{x}$ that minimizes $(x_1 + 1)^2 + x_2^2$ while keeping x_1 less than or equal to 1.

(a) Write this optimization problem in standard form, and clearly specify the objective function $f(\vec{x})$ and the constraint functions $g(\vec{x})$ and/or $h(\vec{x})$.

(b) Consider two candidate solutions $\vec{x}_A = [1,\ 0]^T$ and $\vec{x}_B = [-1,\ 0]^T$. Which of these two points satisfies all of the KKT conditions, or do both or neither? If a point satisfies all the applicable conditions, demonstrate that each is satisfied. If a point does not satisfy all the conditions, indicate which conditions or combination of conditions cannot be satisfied.

Question 4. (10 points) Data sets A and B below were generated from the same line with different amounts of noise. Panes A and B show least-squares linear fits.



Which linear fit, A or B, has the higher r^2 value? Justify your answer.

Question 5. (10 points) The polynomial regression code below has initialized the Z matrix needed to solve the normal equations. Write code to fill in this matrix with the appropriate values for polynomial regression. `x` and `y` are column vectors containing the n data points and `m` is the order of the polynomial $a_0 + a_1 x + \ldots + a_m x^m$.

```
function a = polynomial_regression(x, y, m)
    % Fit a polynomial of order m to data by least squares.
    % x, y: column vectors of the n data points
    % m: order of the polynomial
    % a: column vector of coefficients [a0; a1; ...; am]

    n = length(x);
    Z = zeros(n, m+1);

    % fill in Z here









    a =(Z'*Z)\(Z'*y);
end
```

Solutions

Question 1

(a) $\nabla f(\vec{x}) = [4x_1,\ 2x_2]^T$, so $\nabla f(\vec{x}_0) = [4,\ 4]^T$ and $\vec{p}_0 = -\nabla f(\vec{x}_0) = [-4,\ -4]^T$.

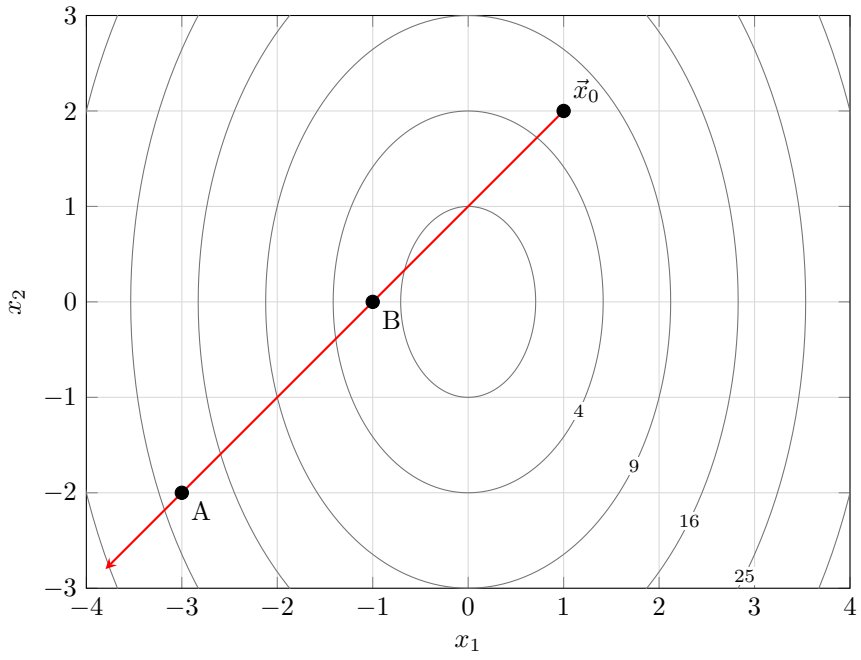
(b) See the plot below. *Note (not required):* the line is perpendicular to the contour through $\vec{x}_0$.

(c) A: $\vec{x}_0 + 1\,\vec{p}_0 = [-3,\ -2]^T$. B: $\vec{x}_0 + 0.5\,\vec{p}_0 = [-1,\ 0]^T$.

(d) The Armijo condition is $f(\vec{x}_0 + \alpha\,\vec{p}_0) \le f(\vec{x}_0) + \mu_1\,\alpha\,\nabla f(\vec{x}_0)^T\vec{p}_0$, with $f(\vec{x}_0) = 2+4 = 6$ and $\nabla f(\vec{x}_0)^T\vec{p}_0 = -32$.

A: $f = 2(9) + 4 = 22 > 6 - 10^{-4}(1)(32) = 5.9968$, so A does **not** satisfy it.

B: $f = 2(1) + 0 = 2 \le 6 - 10^{-4}(0.5)(32) = 5.9984$, so **B satisfies** it.



Question 2

(a) All of the operations are elementwise:

```
x = x - alpha./(epsilon + sqrt(s_hat)).*v_hat;
```

(b) The function takes the *gradient*, not f, so the script must define $\nabla f(\vec{x}) = [4x_1,\ 2x_2]^T$ as a function of a column vector:

```
grad_f = @(x) [4*x(1); 2*x(2)];
xstar = adam(grad_f, [1; 2], 0.3, 0.9, 0.999, 1e-8, 1e-6)
```

(c) **Path 2.**

Step by step detailed justification:

1. Since this is a quadratic function, Newton's method will solve it in a single step.
2. Path 1 goes straight to $[0,\ 0]^T$ in one step.
3. Therefore, by 1 and 2, path 1 is Newton's method.
4. Gradient descent: $\vec{x}_{k+1} = \vec{x}_k - \alpha \nabla f(\vec{x}_k)$, and $\nabla f(\vec{x}_k)$ is perpendicular to the contour through $\vec{x}_k$
5. Every step of path 3 is perpendicular to the contour it starts on.
6. Therefore, by 4 and 5, path 3 is gradient descent.
7. Therefore, by 3 and 6, the remaining path, path 2, is Adam.

Note (not required): the momentum $\vec{v}$ carries Adam past the minimum, which causes the overshoot. Also, working out Newton's single step: $\vec{p}_0 = -H(\vec{x}_0) \backslash \nabla f(\vec{x}_0) = -\begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \backslash \begin{bmatrix} 4 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$, so $\vec{x}_1 = \vec{x}_0 + \vec{p}_0 = [0,\ 0]^T$

Question 3

(a) $x_1 \leq 1 \Rightarrow x_1 - 1 \leq 0$, so the standard form is

$$\begin{aligned} \underset{\vec{x}}{\text{minimize}} \quad & f(\vec{x}) = (x_1 + 1)^2 + x_2^2 \\ \text{subject to} \quad & g(\vec{x}) = x_1 - 1 \leq 0 \end{aligned}$$

There are no equality constraints, so there is no $h(\vec{x})$.

(b) Only $\vec{x}_B$. With $\nabla f = [2(x_1 + 1),\ 2x_2]^T$ and $\nabla g = [1,\ 0]^T$, the conditions are stationarity $\nabla f(\vec{x}^*) + \sigma^* \nabla g(\vec{x}^*) = \vec{0}$, feasibility $g(\vec{x}^*) \leq 0$, $\sigma^* \geq 0$, and complementary slackness $\sigma^* g(\vec{x}^*) = 0$. (The condition $\vec{h}(\vec{x}^*) = \vec{0}$ does not apply.)

	$\vec{x}_A = [1,\ 0]^T$	$\vec{x}_B = [-1,\ 0]^T$
Stationarity	$[4,\ 0]^T + \sigma[1,\ 0]^T = \vec{0} \Rightarrow \sigma = -4$	$[0,\ 0]^T + \sigma[1,\ 0]^T = \vec{0} \Rightarrow \sigma^* = 0$
$g \leq 0$	$g = 0$, satisfied (active)	$g = -2$, satisfied (inactive)
$\sigma \geq 0$	$-4 < 0$, **violated**	satisfied
$\sigma g = 0$	satisfied	satisfied

Note (not required): $\vec{x}_B$ is the unconstrained minimum and it is feasible, so the constraint is inactive and its multiplier is zero. At $\vec{x}_A$ the negative multiplier means that f decreases when moving off the constraint into the feasible region.

Question 4

A.

Step by step detailed justification:

1. $r^2 = \dfrac{S_t - S_r}{S_t}, \quad S_r = \sum (y_i - \hat{y}_i)^2, \quad S_t = \sum (y_i - \bar{y})^2$
2. The points in A are much closer to the line than in B, so S_r is smaller for A.
3. The spread of y about its mean is similar in A and B, so S_t is similar.
4. Therefore, by 1 through 3, r^2 is higher for A.

Note (not required): the computed values are $r^2 = 0.99$ for A and 0.59 for B.

Question 5

Row i of Z is $[1,\ x_i,\ x_i^2,\ \ldots,\ x_i^m]$, that is, $z_{ji} = x_i^j$ for $j = 0, \ldots, m$. Matlab indices start at 1, so the column for power j is column $j+1$:

```
for i = 1:n
    for j = 0:m
        Z(i, j+1) = x(i)^j;
    end
end
```

A single loop over j with `Z(:, j+1) = x.^j` is also fine.